

StormWatch Weather Data and MCP Tool Contracts

Document ID: SW-003 Category: data-and-tools Version: 1.0 Status: Synthetic workshop reference Security classification: workshop-public

Data Acquisition

Live mode resolves a user-supplied city with OpenWeather Direct Geocoding before requesting the five-day, three-hour forecast by latitude and longitude. Requests use HTTPS, metric units, cancellation, and a finite timeout.

Fixture mode reads the supplied synthetic JSON file and makes no live request. It is the preferred workshop validation path because it is deterministic and does not require an OpenWeather key.

The API key is read only from OPENWEATHER_API_KEY in live mode. It must not appear in source, output, logs, exceptions, returned URLs, MCP responses, or browser content.

Domain Records

Location

Field	Type	Meaning
Name	string	Resolved city name
Country	string	Country code
Latitude	number	Decimal latitude
Longitude	number	Decimal longitude
State	optional string	Resolved state or region
DisplayName	derived string	Non-empty name, state, and country joined for display

ForecastPoint

Field	Unit	Meaning
Timestamp	UTC date-time	Start time of the forecast period
TemperatureC	degrees Celsius	Forecast temperature
WeatherId	OpenWeather condition ID	Numeric weather classification
Description	text	Human-readable weather description
WindSpeedMps	meters per second	Forecast wind speed
Rain3hMm	millimeters per three hours	Rain accumulated in the period
PressureHpa	hectopascals	Forecast atmospheric pressure

If the OpenWeather payload omits the three-hour rain field, StormWatch maps it to 0.0 mm/3h. Missing rain does not mean missing forecast data; it means no reported three-hour rain amount for that point.

Forecast and StormAssessment

A Forecast combines one resolved Location with a non-empty ordered collection of points. A StormAssessment contains a level, capped score, peak forecast point, and evidence reasons.

Display Units

StormWatch scoring and structured output use metric units:

- temperature: degrees Celsius;
- wind speed: meters per second (m/s);
- rain: millimeters per three hours (mm/3h);
- pressure: hectopascals (hPa); and
- time: UTC with an explicit offset or UTC label.

A future interface may display converted values, but scoring must remain metric and the original units must be unambiguous.

MCP Tool Inventory

The stdio server advertises exactly two tools.

get_forecast

Purpose: Get the next five OpenWeather forecast points for a city.

Input:

Parameter	Type	Required	Description
city	string	yes	City name, optionally followed by a country code

Structured result:

Field	Meaning
location	Resolved display name
coordinates	Latitude and longitude
source	OpenWeather 5 Day / 3 Hour Forecast or synthetic fixture
units	Labels for temperature, wind, rain, and pressure
periods	Exactly the next five typed forecast periods

Each period contains UTC time, temperature in Celsius, description, weather ID, wind in m/s, rain in mm/3h, and pressure in hPa.

Use get_forecast when the user asks what the next periods contain but does not need the educational score.

assess_storm_risk

Purpose: Assess the educational storm-risk signal for a city's forecast.

Input:

Parameter	Type	Required	Description
city	string	yes	City name, optionally followed by a country code

Structured result:

Field	Meaning
location	Resolved display name
source	Live OpenWeather or synthetic fixture
level	low, watch, or warning
score	Integer from 0 through 100
peakTimeUtc	Selected peak timestamp
evidence	Every reason that contributed points
disclaimer	Educational-use and non-official-warning qualification

Use assess_storm_risk when the user asks for risk, score, peak, evidence, or an explanation of why a period was selected. The tool composes the existing weather and risk services; it does not duplicate their logic.

RAG and Tool Routing

User need | Retrieval | MCP tool

Explain the 990 hPa threshold | Risk scoring PDF | none

Show the next five periods for Oslo | Optional field definitions | get_forecast

Assess risk for Bengaluru | Responsible-use guidance | assess_storm_risk

Explain a returned score | Risk scoring PDF | assess_storm_risk

Describe MCP architecture | System overview PDF | none

Ask whether an official alert exists | Responsible-use PDF | none; direct user to official source

The PDF corpus contains no current forecast. A city name in a scenario does not authorize the chatbot to skip an MCP call.

Error Contract

Expected failures include blank city input, unknown city, unsuccessful HTTP response, timeout, cancellation, invalid JSON, missing fixture, unreadable fixture, and missing live-mode configuration.

For a normal user-facing failure, the chatbot should:

1. say that StormWatch could not load or assess the requested data;
2. provide a concise, non-secret reason when available;
3. suggest correcting the city, checking configuration, or retrying later;
4. avoid a stack trace or credential-bearing URL; and
5. never manufacture forecast points or a score.

Cancellation should propagate rather than becoming a successful empty result. Unit tests and workshop evaluation must not call live services.

Concision and Determinism

Tool responses are structured rather than preformatted reports. They should contain only the fields needed by the caller, use invariant numeric and timestamp formats, and never write normal application output to stdio. These properties let both GitHub Copilot and the local Foundry client inspect the same contract reliably.